

CURRICULUM MODULE | 2026

GRADE SEVEN

Open topic, mostly in class.

First semester | OAS State Science Day pathway



GRADE BAND PHILOSOPHY

Seventh graders choose their own topic, which gives them the motivation to follow through on a longer process. The structure of the curriculum keeps most of the work happening in class, where teachers can monitor progress and give real-time feedback. Experimentation may happen at home for projects that require time-based observation, but in-class data collection is strongly preferred wherever possible.

The key shift from grades 5 and 6 is that students become responsible for their own question and design. Teachers move from directing to coaching. The milestone structure and journal requirements keep students from falling behind without the teacher managing every step.

THE WHAT IS THE POINT QUESTION

Seventh graders need to know why their project matters before they will commit to it. Build this into topic selection. A student testing whether a sports drink grows plants should be pushed toward a stronger framing: what does that tell us about additives in water, and what agricultural or environmental problem does it connect to? The experiment becomes compelling when a student can answer: if my hypothesis is correct, why should anyone care?

TOPICS THAT HAVE WORKED WELL

Temperature effect on pickleball bounce, surface type and traction testing, natural water additives and plant growth, noise level and reading comprehension, sleep duration and reaction time. These are testable, relevant, and completable in a semester.

SEMESTER 1 TIMELINE

WEEK	PHASE	PRIMARY CLASSROOM	WHAT STUDENTS DO
1	Topic Exploration	Science / History	Brainstorm freely, identify a real-world problem the project connects to, begin the journal
2	Testable Question	Science	Narrow the topic to a testable question; identify independent and dependent variables
3	Background Research	ELA	Read two credible sources; complete a source summary sheet; identify what is already known
4	Hypothesis	Science	Write an if, then, because hypothesis; peer review with a partner
5	Experimental Design	Science	Identify all three variable types; write the procedure, minimum eight steps and three trials
6	Materials and Feasibility	Science / home prep	Complete the materials list; flag anything unavailable; teacher approves the design
7 to 9	Data Collection	Science or home	Run the experiment; record every trial in the journal; note unexpected observations
10	Data Analysis	Math	Calculate averages; construct an appropriate graph; write a pattern description
11	Conclusion	ELA / Science	Write a CER conclusion; draft a short research report
12	Display Board	Art / Science	Design and assemble the board; apply the Display Board Checklist
13	Presentation Prep	ELA	Draft the oral presentation outline; practice with a partner; receive feedback
14	Practice Fair	Science / ELA	Present to a small group with the teacher as judge; respond to questions; self-assess

JOURNAL CHECKS

Collect journals at the end of weeks 3, 6, 9, and 12.

TEACHER NOTES

Hypothesis flexibility

Students should know from the start that they may revise their hypothesis before data collection begins if the design reveals a flaw. This is not a failure, it is how science works. Build a formal hypothesis check into the design approval step.

Design flexibility

Students may adjust the experimental design during data collection if a procedure step is producing unreliable results. Changes must be documented in the journal with a dated explanation.

Everyday materials are enough

Students do not need lab equipment. Rubber bands, cardboard, measuring cups, pharmacy thermometers, and kitchen scales produce valid data for most middle school projects. Say this clearly to students who feel their project is not scientific enough without expensive tools.

History connection

The topic exploration week in History class grounds the experiment in real-world context. A student studying plant growth connects to agricultural history and food deserts. A student studying friction connects to road safety and infrastructure. This is where the cross-curricular framework earns its value.

Elective connections

Music projects covering tempo and mood, frequency and resonance, or rhythm and reaction time can involve the music teacher. Surface and traction projects can involve the PE teacher. Do not force these connections, but note them when they arise naturally.

STANDARDS ADDRESSED

OHIO LEARNING STANDARDS FOR SCIENCE, OLS 2018

- All eight inquiry abilities for grades 5 to 8
- Grade 7 content connections in ecosystems, physical science, and Earth science align with OLS content standards depending on the project selected

CCSS ELA AND LITERACY IN SCIENCE

- RST.6–8.1, RST.6–8.2, RST.6–8.8: reading and evaluating science sources
- WHST.6–8.1: argumentative writing through a claim, evidence, reasoning conclusion
- WHST.6–8.2: informative writing through the procedure and methods section
- WHST.6–8.7 and WHST.6–8.8: conducting research, assessing source credibility, citing sources
- SL.6–8.4 and SL.6–8.5: oral presentation including a visual display

CCSS MATHEMATICS

- 6.SP.A.1 to 3 and 6.SP.B.4 to 5: statistical questions, data displays, measures of center
- 7.SP.A.1 to 2: sampling and inference from data
- 7.SP.B.3: comparing distributions from multiple trial sets